

DR. JANE A. DOE

Computational Neuroscientist · Postdoctoral Researcher

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🌐 janedoe

RESEARCH INTERESTS

Neural computation and learning dynamics, biologically plausible deep learning, large-scale analysis of electrophysiological data, and the development of open, reproducible tools for computational neuroscience.

EDUCATION

Ph.D. in Computational Neuroscience

2016 – 2021

University of Oslo, Norway

Thesis: *Predictive coding in recurrent cortical circuits*. Advisor: Prof. Erik Larsen. Awarded with distinction.

M.Sc. in Physics

2014 – 2016

ETH Zürich, Switzerland

Specialisation in complex systems and statistical mechanics.

B.Sc. in Mathematics

2011 – 2014

University of Cambridge, United Kingdom

ACADEMIC APPOINTMENTS

Postdoctoral Researcher

2021 – Present

Simula Research Laboratory, Oslo

- Lead a project on scalable inference methods for spiking neural network models, funded by a Research Council of Norway grant.
- Supervise two Ph.D. students and mentor three M.Sc. thesis projects.

Visiting Researcher

Summer 2019

Allen Institute, Seattle, USA

- Developed analysis pipelines for large-scale two-photon imaging data, now part of the institute's open toolkit.

SELECTED PUBLICATIONS

JOURNAL ARTICLES

- [1] **Doe, J.**, Larsen, E., & Smith, R. (2023). Predictive coding without backpropagation. *Nature Neuroscience*, 26(4), 512–524.
- [2] Smith, R., **Doe, J.**, & Kaur, P. (2022). Scalable inference for spiking networks. *PLOS Computational Biology*, 18(9), e1010422.

CONFERENCE PAPERS

- [1] **Doe, J.** & Larsen, E. (2021). Learning dynamics in recurrent cortical models. In *Advances in Neural Information Processing Systems (NeurIPS)*.

TEACHING EXPERIENCE

Instructor – Introduction to Computational Neuroscience

2022, 2023

University of Oslo

Designed and taught a 30-student graduate seminar; median evaluation 4.8/5.

Teaching Assistant – Machine Learning

2018 – 2020

University of Oslo

Led weekly lab sessions and coursework for approximately 120 students.

GRANTS, AWARDS & HONORS

- | | |
|------|---|
| 2022 | Research Council of Norway Mobility Grant (NOK 1.2M). |
| 2021 | Best Dissertation Award, Faculty of Mathematics and Natural Sciences. |
| 2019 | NeurIPS Travel Award. |
| 2016 | ETH Excellence Scholarship. |

INVITED TALKS & PRESENTATIONS

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| 2023 | “Learning without backpropagation”, Cosyne, Lisbon (invited talk). |
| 2022 | “Open tools for neural data”, FENS Forum, Paris (poster). |

TECHNICAL SKILLS

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| Programming | Python, Julia, C++, R, Bash |
| Scientific Computing | PyTorch, JAX, NumPy/SciPy, MPI, CUDA |
| Tools & Practice | Git, Docker, L ^A T _E X, reproducible pipelines, HPC clusters |

LANGUAGES

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|------------------|-------------------|
| English | Native |
| Norwegian | Fluent (C1) |
| German | Intermediate (B1) |

PROFESSIONAL SERVICE

- Reviewer: *Nature Neuroscience*, *NeurIPS*, *PLOS Computational Biology*.
- Co-organiser, Oslo Computational Neuroscience Seminar (2022 – present).

REFERENCES

Prof. Erik Larsen

Department of Informatics, University of Oslo

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Dr. Priya Kaur

Group Leader, Simula Research Laboratory

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